Conestoga College

Institute of Technology and Advanced Learning

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Revision History

Revision	Description of Change	Date
1.0	Updated 2019-2020 version according to the course review at the end of academic year 2018-2019 (Monzur Kabir)	Jan 9, 2020
2.0	Updated 2020-2021 version for remote delivery (Alex Tugulea)	Jan 9, 2021
3.0	Updated 2025-2026 version for in-person delivery (So-Ra Chung)	Jan 12, 2026

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ESE Capstone Project Guidelines

1. Introduction

This document provides a set of guidelines for students to use while planning and implementing their Electronic Systems Engineering (ESE) Capstone Project.

Capstone project consists of two 4th year courses:

1. Capstone I (Capstone, Part 1)
2. Capstone II (Capstone, Part 2)

Capstone project is about practicing design and development of a solution to a complex engineering problem. Students will practice engineering design and development activities including electronic design, software development, project management as well as applications of engineering process that takes safety, environmental and societal impact into consideration.

1.1. Objectives

The objective of the ESE Capstone Project is to prepare students for engineering practice through a major design experience based on the knowledge and skills acquired in earlier program work.

"*3.4.4.4 The engineering curriculum must culminate in a significant design experience conducted under the professional responsibility of faculty licensed to practice engineering in Canada, preferably in the jurisdiction in which the institution is located. The significant design experience is based on the knowledge and skills acquired in earlier work and it preferably gives students an involvement in team work and project management."*

> *~ CEAB Accreditation Criteria and Procedure, 2017*
> https://engineerscanada.ca/sites/default/files/accreditation-criteria-procedures-2017.pdf

The project must be about design and development of a solution to a complex engineering problem.

> *"In 2012, the CEAB adopted the definition of "complex problem" used in the Washington Accord (WA) exemplar of graduate attributes. A defining characteristic of the engineering profession is the ability to work with complexity and uncertainty given that all real engineering projects are different from one another. Accordingly, the notion of complex engineering problems and the solving of complex problem are central to the definition of certain attributes.*
> *According to the CEAB, a complex engineering problem is defined by the following characteristics:*
>
> 1. *It must require the application of in-depth knowledge*
> 2. *It must satisfy at least one of the following additional characteristics:*
>   - *involves wide-ranging or conflicting issues*
>   - *has no obvious solution such that originality is required*
>   - *involves infrequently encountered issues*
>   - *is outside accepted standards and codes*
>   - *involves diverse stakeholders and needs*
>   - *is posed at a high-level with many components or sub-problems "*

~ CEAB Guide to Outcomes-Based Criteria, 2015
https://engineerscanada.ca/sites/default/files/draft_program_visitor_guide_v1.25.pdf

The capstone project incorporates engineering standards and realistic constraints that include most of the following considerations:

- economic
- environmental
- sustainability
- manufacturability
- resource management
- ethical
- health and safety
- social
- achievability
- marketability

A number of CEAB graduate attributes are associated with the capstone project (see Appendix E, which also provides the complete list of CEAB graduate attributes.

1.2. Academic Outcomes

See the course learning outcomes in the course outlines

1.3. Project Outcomes and Skill Development

Project Outcome

Upon successful completion of the capstone project, each team will deliver a fully functional integrated electronic system prototype that consists of significant electronic hardware and software components of the team's own design.

The system must provide a solution to an authentic real-world problem with societal, industrial and/or economic implications.

Skill Development

The capstone project is intended to emulate an industry project in its design and implementation lifecycle. Throughout the project work, students will practice:

- Engineering process of new product development - investigation/research, engineering design, product prototyping, test-plan development, verification & validation tests, and performance analysis.
- Lab safety procedure, safety aspects of an engineering design & product implementation, and assessment of adverse societal/environmental impact.
- Project management, professionalism, work ethics and lifelong learning.
- Communication skills and salesmanship through oral presentation, practical demonstration and technical reports.

2. Project Topic and Team Related Guidelines

Project Topic Related Guidelines

Each project should include both electronic hardware and software design/implementation at an engineering level.

Each project should have an industry sponsor. In case of no sponsor, students should consult with knowledgeable professionals in the field related to their project in order to make sure that their project is professionally meaningful. For example, if one is going to do something for the blind, should consult with a professional association like CNIB (Canadian National Institute for the Blind).

Any research projects that involve human testing must be approved by the College's Research Ethics Board. Depending on the nature of the proposed project, teams may be required to apply for project approval prior to proceeding with any further project work. Approval must be received before any involvement of human subject occurs.

Project Team Related Guidelines

Suggested team size is between two and three members - bigger or smaller teams are not permitted under normal circumstances. The faculty team approves the teams at the beginning of the capstone project. There will be no change in team in normal circumstance. The faculty team preserve the right to disallow a particular team and may assign students into other teams.

In case of a team larger than three members, the team must be divided into two or more distinct sub-groups of two or three members. Each sub-group will be treated like a separate team in most of the time for the purpose of performance evaluation.

Every team member must be responsible for engineering investigation, design and prototyping. Each member is also responsible for contributing to all documentations. If faculty team determines that a team member is not doing his/her fair share of work, or is significantly underperforming, faculty team has the right to evaluate the team members separately based on their individual contributions.

Students' progress will be reviewed in weekly basis as indicated section 3. If the faculty team determines that a team or team member is significantly underperforming, the faculty team may assign a faculty member for special mentoring.

3. Project Milestones and Evaluation

3.1. Purpose, Outcomes and Associated Grad-Attributes

Milestone	Purpose	Grad-Attributes
Capstone 1		
Project Proposal Documents and Presentation	To propose a project for faculty-team approval to proceed.	• Investigation (literature review) • Impact of engineering on society & environment • Life-long learning (identifying learning needs and acquiring the skills/knowledge) • Communication skills
User Requirements and Project Plan	• To record all user requirements. This will be the basis for the system validation test at the end of your project. • To breakdown the project work into smaller tasks, set milestones and deadlines. • To plan project work and assign resources for each task. This plan will be the basis of project management performance evaluation. • To analyze risk and write a risk management plan	• Problem Analysis • Engineering Design (user requirement) • Project Management
Investigation, Research, Experiment	To perform investigation, research, experiment for the purpose of engineering design decisions.	• Problem Analysis • Investigation (research, experimentation, simulation) • Engineering Design (hardware/software) • Use of Engineering Tools
Engineering Design & Test-Plan document, and Presentation & Demonstration	• To document design decisions with justifications, design work (schematic-diagram, flow-chart, 3D-drawing and mathematical algorithm, for example) • To perform verification test on individual functional units independently as well as to perform verification & validation test on the integrated system • To re-assess risk and re-write a risk management plan	• Engineering Design (hardware/software) • Project Management (risk re-assessment and mitigation plan) • Communication Skills

Note: each milestone also associates three more graduate attributes: **Project Management**, **Individual & Teamwork**, and **Professionalism**

Milestone	Purpose	Grad-Attributes
Capstone 2		
Prototyping, Testing and Performance Analysis	• To prototype each component as per design, and perform verification test on the prototype as per test plans • To integrate all components to complete the system prototype, and perform verification & validation test on the integrated system as per test plan • To record all test results, and perform analysis to assess the performance of individual functional units as well as the integrated system	• Problem Analysis (analysis of test results) • Engineering Design (prototyping, testing, design refinement/change as needed) • Investigation (research and experimentation, as needed) • Use of Engineering Tools
Final Report, Presentation and Demonstration	• To document all significant work in a single place (most of the components will come from the proposal document, design & test-plan document plus outcomes of previous milestone Prototyping, Testing and Performance Analysis) • To comment on the project success, strengths & weaknesses, and learning experience • To make some recommendations for future students who may undertake similar project or want to continue the same project • To present and demonstrate the project work to wider audience (students, faculty & staff, sponsors and guests)	• Engineering Design (hardware/software design, evidence of prototyping and test results) • Problem Analysis (analysis of test results) • Communication Skills

Note: each milestone also associates three more graduate attributes: **Project Management**, **Individual & Teamwork**, and **Professionalism**

3.2. Deliverables, Deadlines and Marks Distribution

Each submission must be in the designated eConestoga drop-box. The deadline for each submission is on Wednesdays 10 PM before the Thursday 8 AM debriefing day, unless specified otherwise.

Capstone 1 (Semester 7)		Proposal	Project Management	User Requirements and project Plan	Investigation and Design	Presentation & Demo	Design Document and Test Plan
Week #	**Activity/Outcome/Evaluation**	**Marks**					
1	• (Monday) Self-Briefing – Student Resources • (Thursday) Debriefing - Project Outline (review and feedback), Project Guidelines						
2	• (Thursday) Debriefing - Proposal (progress review and mentoring)						
3	• (Thursday) Debriefing - Proposal (review of proposal draft and feedback)						
4	• (Thursday) Oral Presentation - Proposal • (Thursday) Report Submission - Proposal • (Thursday) Evaluation - Project Management	5% 15%	4%				
5	• (Thursday) Debriefing - User Requirements and Project Plan (progress review and mentoring)						
6	• (Thursday) Document Submission and Debriefing - User Req. and Project Plan • (Thursday) Evaluation - Project Management		3%	10%			
7	• (Thursday) Debriefing - Investigation and Design (progress review and mentoring)						
WEEK8	Student Success Week						
9	• (Thursday) Debriefing - Investigation and Design (progress review and mentoring)						
10	• (Thursday) Oral Assessment - Investigation and Design (**Assessment 1)** • (Thursday) Evaluation - Project Management		4%		10%		
11 to 13	• (Thursday) Debriefing - Investigation and Design (progress review and mentoring)						
14	• (Thursday) Oral Presentation and Demonstration - Investigation, Design and Test-Plan • (Thursday) Report Submission – Engineering Design and Test-Plan • (Thursday) Oral Assessment - Investigation and Design (**Assessment 2)** • (Thursday) Evaluation - Project Management		4%		15%	10%	20%
15	Final Exam Week						
S	Total	20%	15%	10%	25%	10%	20%

Capstone 2 (Semester 8)		Prototype Dev. and Testing	Project Management and Professionalism	System Integration and Performance Analysis	Presentation and Demonstration	Final Report
Week#	**Activity/Outcome/Evaluation**	**Marks**				
1 (TBA)	• (Monday) Briefing					
2 (TBA)	• Victoria Day (no debriefing)					
3 (TBA)	• (Monday) Debriefing – Review of updated User Requirements and Project Plan		2.5%			
4 (TBA)	• (Monday) Oral Assessment - Engineering Dev & Testing (**Assessment 1)** • (Monday) Evaluation - Project Management	8%	2.5%			
5 (TBA)	• (Monday) Debriefing - Engineering Dev & Testing (progress review and mentoring)					
6 (TBA)	• (Monday) Debriefing - Engineering Dev & Testing (progress review and mentoring)					
7 (TBA)	• (Monday) Oral Assessment - Engineering Dev & Testing (**Assessment 2)** • (Monday) Evaluation - Project Management	10%	2.5%			
Student Success Week (TBA)						
8 (TBA)	• Debriefing					
9 (TBA)	• (Monday) Briefing - System Integration and Performance Analysis • (Monday) Oral Assessment - Engineering Dev and Testing (**Assessment 3)** • (Monday) Evaluation - Project Management	12%	2.5%			
10 (TBA)	• (Monday) Debriefing - System Integration and Performance Analysis (progress review and mentoring)					
11 (TBA)	• (Monday) Debriefing - System Integration and Performance Analysis (progress review and mentoring)					
12 (TBA)	• (Monday) Oral Assessment - - System Integration and Performance Analysis (**Last Technical Assessment**) • (Monday) Evaluation - Project Management		2.5%	15%		
13 (TBA)	• (Monday, Aug 16) Oral Presentation and Demonstration – Project Final • (Monday, Aug 16) Exit interview, end-of-semester survey, feedback on final report draft • (Thursday, August 20) Final Report submission • (Thursday, August 20) Evaluation - Project Management		2.5%		15%	25%
14 (TBA)	• Tech Showcase (nominated team only) (no marks)					

3.3. Evaluation

The table below gives the list of assessments. All assessments will use rubrics (separately published) and prescribed guidelines (written in this document)

- Marks in rubrics reflect how a project-team is performing. These marks are not necessarily the marks for individual students. Individual students' marks will be based on the rubrics mark and the performance differential between individual students of a team.
- Written deliverables must be submitted before oral-assessment/debriefing/presentation or deadlines, as applicable. Otherwise, up to 20% may be deducted from what obtained on the rubrics.
- Each phase of the capstone project (semester 7 and 8) includes a **safety quiz**. This is a **Pass/Fail** evaluation, and students must pass the quiz before using lab equipment for any courses taken in either semester.

Milestone	Basis of Evaluation
Capstone 1	
Project Proposal Doc. and Presentation	• Oral presentation as per guidelines • Written document as per guideline and document structure
User Req and Project Plan	• Written document as per guideline and document structure
Investigation, Research, Experiment	• Oral assessment as per guidelines during debriefing sessions (written evidence/log must be produced to support your claims) • Successful demonstration as per guidelines during debriefing sessions
Design & Test-Plan, and Presentation & Demonstration	• Oral presentation as per guidelines • Successful demonstration as per guidelines during debriefing sessions • Written document as per guidelines and document structure
Project Management, Teamwork and Professionalism	• Project management: project status versus project plan • Teamwork and professionalism: based on observations made by faculty members during project activities including debriefing sessions (see the relevant section of this guideline for detail)
Capstone 2	
Updated User Req. and Project Plan	• Written document as per guideline and document structure
Prototyping, Testing and Performance Analysis	• Oral assessment as per guidelines during debriefing sessions (written evidence/log must be produced to support your claims) • Successful demonstration as per guidelines during debriefing sessions
Final Report, Presentation and Demonstration	• Oral presentation as per guidelines • Successful demonstration as per guidelines • Written document as per guidelines and document structure

3.4. Debriefing

There will be weekly debriefing session for each project team. There are two different types of debriefing sessions:

- Progress review and mentoring
- Oral assessment and demonstration

All debriefing sessions (both types) are associated with performance evaluation of project management, teamwork and professionalism. Students are expected to:

- Be ready for debriefing with printed status report and project binder.
- Be ready for a demo, if any (keep relevant document such as circuit/network/schematic diagram handy).
- Upload all documents in designated eConestoga drop-box before the debriefing unless there is any explicitly given submission deadline.
- Use debriefing time effectively.
- Team members should be on the same page. Each team member should know each other's work and status and be able to represent another member in her/his absence.
- Appropriate attire and communication etiquette.
- Produce up-to-date project binder that includes
  - Weekly project status report
  - Engineering log
  - Record of meeting minutes/notes
  - Any other documents produced by the team
  - Datasheets and relevant articles/technical documents

4. Project Deliverables Description – Part 1 (Semester 7)

Each student shall use industry standard techniques and best practices for all design, fabrication, and testing. An engineering logbook shall be used to document all work including the investigation, design, development, and testing of all systems and subsystems. Project management tools will be used to control the project.

The capstone project is intended to emulate an industry project in its design and implementation lifecycle. Any documents described in this section must be submitted using the standard ESE format specified in Appendix A. The evaluation rubrics for the project deliverables are available in a separate document.

4.1. Project Proposal

Deliverables:

- Project Idea document
- Project Outline document
- Project Proposal document
- Oral presentation of the project proposal

Project idea and outline are two initial phases towards a complete project proposal. Detail of these phases and the structure of the documents are given in the following sub-sections.

4.1.1. Project Idea (before formally starting the capstone project)

Students are expected to brainstorm for developing a project idea, share their project idea with the designated professor(s) before formally starting the capstone project in semester 7. This phase is designed to get an early feedback from the professors. Students will try to find an industry sponsor as well in this phase.

Guidelines:

- Project teams should have 2 or 3 members.
- Project must include engineering design work that a team will carry out.
- Design work should include both electronic hardware and software design.
- All members of a team must carry out a part of engineering design.
- Students should strive to have a context, which can be societal, industrial (business or entrepreneur) and/or economic value.
- Students should consult with knowledgeable professionals in the field related to their project to make sure that their project is professionally meaningful. For example, if they are going to do something for blind people, they should consult with a professional association like CNIB (Canadian National Institute for the Blind).

Structure of the Project Idea document

- Project title
- Name of team members
- Project justification (why is needed?)
- UN Sustainable Development Goal (which ones are covered?)
- Target beneficiary group (who needs it? Have you considered the “Design Justice”?)
- Project description (what needs to be done?)
- Anticipated budget (does it cover the “circular economy”?)
- Sponsors or professional supporters

4.1.2. Project Outline

Project outline is a refined and extended form of the project idea. Students will write this document in accordance with the document structure given below.

Structure of the Project Outline document

- **Cover Page**: include the document name 'Capstone Project Outline', project title, student names & IDs, academic term/year, course title and code, program name and college name.
- **Background**:
  - Problem/need statement: state what problem to be solved.
  - Rationale: justify why this problem needs to be solved.
- **Project Description**: describe the proposed solution to the problem.
- **Literature Review**: identify current solutions for this problem (citation required).
- **Technical level and scope**: make a case that the project work is 4th year engineering level, and the project scope is appropriate for a two-semester-long capstone project.
- **Project Context**: identify project context as adding societal, industrial (business or entrepreneur) and/or economic value.
- **Reference and Bibliography**: see appendix B for the format

4.1.3. Project Proposal

Project proposal is the most comprehensive and presentable document produced in the proposal phase. Students will write the project proposal document in accordance with the structure, and also present the proposal orally in accordance with the guideline.

The project proposal must be submitted to the faculty team for approval. The faculty team will assess the feasibility of the proposed project with respect to technical difficulty and complexity, available expertise and resources, cost, etc. Faculty team may ask students to revise the proposal. Once the final approval is received, any significant change to the project will require faculty approval.

Structure of the Project Proposal Document

- **Cover Page**: include the document name 'Capstone Project Proposal', project title, student names & IDs, academic term/year, course title and code, program name and college name.
- **Table of Content** (include page number)
- **Introduction**
  - Purpose of the document: identify the purpose and audience of this document.
  - Project Definition: name and briefly describe the proposed project (*elaborate description to be covered in separate section*).

    Note: project title, definition and description are three levels of detailing. Project definition is expected to be a short paragraph, whereas project *description is more detail including a system block diagram, where the blocks are major system components.*

- Background:
  - Problem/need statement: state what problem to be solved.
  - Rationale: justify why this problem needs to be solved.
- Proposed Solution
  - Describe the solution.
  - Indicate if there is any similar solution available in literature. If available, state how the proposed one is different. Highlight the benefit of the proposed solution.
  - Identify which part of the system the proposed project is focusing on.
  - List the goals the project wants to achieve.

- **Literature Review**

  (C*itation of authoritative sources is required. Authoritative sources include well-known technical/scientific journals, magazines, industry white papers and product descriptions from original vendors*).

  Perform literature review for the following purposes:

  - Establish that the research/project topic is currently a topic of active research and development and has potential to be a practical solution to a real-world problem.
  - Provide the readers of this document with some examples of research/project that are already reported in authoritative journals and technical magazines (this is a sort of environmental/boundary scan)
  - Identify and describe the literatures that form the basis or starting point of this project work (these can be any combination of, for example, a project idea, research work, off-the-shelf components, industry white papers on 'coming soon' products or solutions, new regulations and open standards)
  - Compare the proposed solution to existing solutions, if any.

- **Project Details**
  - Describe in detail what the proposed system will be and/or do (a system block diagram should be used here).
  - Identify if there is anything new/unique/novel in the proposed project.
  - Technical level and scope: make a case that the project work is 4th year engineering level, and the project scope is appropriate for a two-semester-long capstone project.
  - Identify project context as adding societal, industrial (business or entrepreneur) and/or economic value.
  - Identify professional responsibility attribute(s) such as safety, ethics, societal and environmental concern.
  - Impact on society: project should have positive impact on society, demonstrated by one or more of the following:
    - Academic, industrial or professional sponsorship, support or endorsement.
    - Reasoned argument as to the societal relevance of the project

- **Project Feasibility**
    - Expertise
        - identify the knowledge and skills that are required for successful completion of the project
        - identify any knowledge that must be acquired to complete this project
        - identify sources of expertise that are available to help complete this project
        - identify possible external advisors for this project
    - Resources
        - identify the tools and resources that are required for successful completion of the project
        - Specify the estimated cost of the project.
        - Identify sponsors of this project.
    - Budget Requirement

- **Risk Analysis**

  The risk analysis describes which risks might affect the project. It includes:
    - technical, quality and performance risks
    - project management risks
    - organizational risks
    - external risks

  The risk analysis report must include:
    - an evaluation of the impact of each risk on the major project objectives
    - the overall risk for the project
    - a list of prioritized risks

- **Reference and Bibliography:** See appendix B for the format

4.1.4. Project Proposal Presentation

- 15 minutes oral presentation plus 5 minutes Q&A (question & answer).
- All capstone students are required to be the audience of other students' presentations. They are also expected to ask questions during Q&A session.
- External guests may be invited and some of them may participate in the performance evaluation.
- Presentation material must be submitted in the designated eConestoga drop-box prior to the presentation.

Presentation Structure/Sequence

The structure/sequence of the presentation is same as the proposal document structure.

- Introduction (introduce yourself, state the purpose of the presentation, project title and definition)
- Literature Review
- Project Description
- Project Feasibility
- Risks Analysis

Evaluation Criteria

- Communication skills (project related)
    - Each part of the presentation should cover all aspects as listed in the proposal document structure.
    - Quality of each part
- Communication skills (overall)
    - Well integrated and logically sequenced presentation
    - Smooth presentation flow, well-articulated answer to relevant questions, and good aural quality.
    - Clear presentation materials (power point, audio and video, for example) – easy to read, see and hear
- Professionalism (overall): professional attire and behaviour consistent with professional work-ethic (punctuality, timeliness, responsibility, active participation and appropriate communication etiquette)

Further Guidelines:

- The goal of this presentation is to establish that the proposed project meets all the requirements to be considered a capstone project.
- The presentation makes a case that the proposed project is 4th year engineering level, and the defined scope of the project is appropriate for two semester long team project.
- The <u>core of the presentation</u> is the summary of key components of the proposal document. Highlight the key components and be ready to explain/present further details if anyone from the audience asks for it.

- Background information: expect that some of the audience is not familiar with the project, so start with a brief overview. Use generic terms wherever possible or explain the term (example: if you use the term OpenCV, clearly tell what it is at the beginning).

4.2. User Requirements and Project Plan

Deliverables: A document that includes:

- Documented user requirements
- Project plan including work breakdown, resource allocation, risk analysis & risk management plan, and project schedule such as Gantt-chart

In the user requirement phase, a system-designer documents the specifics of the user/client requirements (assuming that system description is already available in the project proposal document). Note that, a 'user' may provide any level of detailing in between very generic to very specific. A more generic requirement gives a designer more flexibility to design but also passes the responsibility to the designer to come up with required level of specifics. No matter who does this (user or designer), the user requirement document should provide some specifics on:

- The scope of the capabilities of the system, which will be designed and developed.
- A set of functional/operational behaviour of the system under a typical operating condition, which has to be specified in the user requirement document as well.

In the project plan phase, a project team breaks the whole system into a number of **functional units**, which forms the whole system when properly integrated. A functional unit meets the following criteria:

- It can be a piece of hardware or software.
- It is an essential system component - either 'to be designed' or off-the-shelf component.
- It should be assignable to a single team member as an independent task.

Note 1: Each student must be responsible for at least one 'to be designed' functional unit that involves 4th year level engineering design and development, and should be big enough for two-semester-long project work. That is, if a team has three students, there must be at least three such functional units specified at this stage of the project.

Note 2: Modularity is expected in system design & development work. It is expected that each team member will work on individual functional units separately as long as it makes sense, and test each unit separately to ensure that it will work properly when integrated with other system components.

A project plan also needs a system integration plan where a set of system integration tasks to be defined. However, students may opt to elaborate this part later (at the beginning of capstone part-2 in semester 8).

Apart from tasks of functional unit development and integration, a project involves selection of off-the-shelf components and may also involve tasks of new resources and knowledge acquirement.

Next step is to assign tasks to individual team members. The task distribution among the team members should be such that the workload (including engineering design work) is evenly distributed.

Next step is scheduling the tasks. Students should identify the most complex, challenging and/or uncertain tasks first, and start working on those tasks as early as possible. They should research, experiment, simulate, acquire knowledge/skills, and also ask for help, as needed, in order to reduce the risk associated with those tasks as early as possible.

Some procurement and outsourcing may cause delay. You should send purchase order out as early as possible. Also, plan your work in such a way that you don't need to sit idle until ordered resources are available.

Last part of the project plan is to perform risks analysis and risk management plan. Assuming that you already performed risk analysis (in the proposal phase), you will refine that. In addition, you will write a risk management plan to define some kind of 'plan B'.

4.2.1. Structure of User Requirements and Project Plan Document

- **Cover Page:** include the document name 'User Requirements and Project Plan', project title, student names & IDs, academic term/year, course title and code, program name and college name.
- **Table of Content** (include page number)
- **Introduction**
  - Purpose of the document: identify the purpose and audience of the document.
  - Project definition and description (*you may fill this sub-section with relevant parts of the Project Proposal document*)

- **User Requirements**
  - Specify the scope of the project using MoSCoW (**Mu**st, **S**hould, **Co**uld, **W**on't) template in appendix C. The scope should be appropriate for a two-semester-long project of a team of 4th year engineering students. Core engineering tasks that you are undertaking must be identified as 'Must'.
  - Specify a set of criteria that the proposed project will meet at its completion. The criteria should be appropriate for a two-semester-long project of a team of 4th year engineering students. Specify the criteria using SMART (**S**pecific, **M**easurable, **A**chievable, **R**ealistic, **T**imely) goal template, given in appendix C.

  Note: MoSCoW scope and SMART criteria together sets the goal the project that has to achieve at the end of the project. At the end of your project you will perform validation tests on the designed and implemented prototype in order to confirm that you achieved those goals.

- **Project Plan**
  - System Breakdown
    - Based on the MoSCoW list, identify off-the-shelf components and 'to be designed' parts of the system.
    - Breakdown of the 'to be designed' parts into a number of **functional units**. These units are expected to be designed, prototyped and tested separately and independently. Note: at least one design and development task for each team member is required.

      (The term 'functional unit' can be a hardware or software, and assignable to a single member of a project team)

  - System Integration
    - At the implementation phase (in Capstone 2) students are expected to perform system integration. At that stage students have to elaborate this

section (for now it is a placeholder for that future task and you may mark it as TBD)

- Resource and Knowledge Requirements
  - List and describe what tools, off-the-shelf components, software required for the project.
  - List and describe what new knowledge and skills you need to acquire to carry out the project. Indicate how you will acquire these knowledge and skills.
  - Update the estimated cost (as in the proposal document) of acquiring resources and knowledge.

- Task Assignment and Schedule
  - Produce a schedule such as Gantt chart that assigns the tasks to individual team members.
  - Sets the milestone(s) and deadline(s) for each task of functional unit design. These milestones and deadlines must be consistent with the assessment/ evaluation dates (see Section 3.2 for detail)

    Note: Each individual student must be responsible for at least one functional unit design and development.

- Risk Analysis and Risk Management Plan
  - Refine the risk analysis, which was initially done in the project proposal.
  - Produce a risk management plan for all major risks.

- Reference and Bibliography: See appendix B for the format.

4.3. Investigation for Engineering Design

Deliverables:

Project binder that includes appropriately documented results of investigation (research, experiment and simulation, for example), engineering design. There is no specific structure suggested for these documents. Refined form of some of these materials is expected to be included in the engineering design document. For that reason, you may use the structure of that document for your work in this phase.

During evaluation of your performance, you are expected to present/demonstrate/defend the following items:

- engineering design alternatives and your design
- justification behind engineering design decisions
- analysis, experiments and simulation
- record of measurement data, experiment/simulation results, and data analysis
- engineering log
- a list of newly acquired (and/or to be acquired) knowledge/skill along with justification why these are required, and status of these acquirements

- an assessment on safety as well as environmental and societal impact the proposed design may have if it is implemented and put into operation
- justifications behind selection of specific off-the-shelf hardware and/or open-source software
- list of incomplete design works, if any, and risk assessment

In this phase, you will start your investigation work for the purpose of engineering design. You will follow the user requirements and project plan to keep your activities on track. For a successful investigation phase, you may consider the following guidelines:

- Explore more than one design solution and be able to defend your design decision.
- Present your engineering design using appropriate means of engineering documentation such as circuit/schematic diagram, flow-chart, state-machine diagram, mathematical formula, and 3D graphics, as appropriate.
- Identify the most complex, challenging and/or uncertain part of the design/development work first, and start working on it as early as possible. Research, prototype, experiment/simulate, acquire knowledge/skill, and also ask for help as needed in order to reduce risk.
- When you select an off-the-shelf hardware and software, be thorough in your selection process and do it based on a sound engineering decision making process.
- List (with short description) the selected key electronic components, open-source software, CAD tools, programming language, communication protocols/systems, mathematical models/algorithms and so on along with a comparative study on alternatives/options and justification behind your selection.
- Be thorough in your engineering investigation:
  - Record key investigation work such as prototype & experiments and simulation.
  - Record test data/results and perform data analysis for the purpose of design decisions.
  - Maintain engineering log.

4.4. Engineering Design and Test Plan

Deliverables: a document that includes:

- Engineering design of all functional units and the integrated system
- Test plans for design verification of each functional unit, and verification & validation of the integrated system

This document is the result of the investigation and design work. Test plans are required for all designed functional units and also for the integrated systems/sub-systems.

Ideally, this document should be written in such a way that another team of students with same background, but not involved in this project so far, can carry out the prototype development work with minimal help from the original designer(s).

Structure of Engineering Design Document and Test Plan

- **Cover Page**: include the document name 'Engineering Design Document and Test Plan', project title, student names & IDs, academic term/year, course title and code, program name and college name.
- **Table of content** (include page number)
- **Introduction**
  - Purpose of the document: identify the purpose and audience of the document.
  - Project Definition: name and briefly describe the proposed project (*elaborate description to be covered in separate section*)
  - System Requirements
    - Scope of the project using MoSCoW template (you may refine the scope defined in the User Requirements document)
    - A set of criteria that the designed and prototyped system must meet. Present it with SMART template (you may refine the criteria defined in the User Requirements document)
- **System Design**
  - System level diagrams & drawings, principle of operation with brief descriptions of each functional unit, indication of what components are 'off-the-shelf' and what are to be designed and developed.
  - Test plans for:
    - verification of system integration
    - validation test to confirm that the system meets the SMART requirement (as defined in User Requirement and Project Plan document)
  - Safety, societal and environmental issues associated with the project and a mitigation plan.
    - Describe how your solution may impact on the environment and society (positive or negative). Describe how to reduce the negative impact, if any.
    - Describe required safety precautions if the product may cause any safety hazard.
  - Brief description/specification of off-the-shelf components (datasheet or detail description of off-the-shelf components such as microcontroller, wireless transceivers, embedded modules, major electrical & electronic components, communication devices, client-server software, open-source software and protocols should be included as an appendix, unless it is too large. If too large, submit those as separate files).

Note: You may create one subsection per functional unit to present your design work. The following sub-section is an example.

- **Design of Functional Units** (multiple sections with appropriate titles – one for each functional unit)
  - Definition of the functional unit.
  - Design specification for the functional unit. Show your design of individual functional unit using circuit/schematic diagram, flow-chart/pseudocode, 2D/3D mechanical drawing, mathematical model/algorithm as appropriate, or any other standard means of presenting engineering design description).
  - Some of the design may involve engineering principles (circuit theory, algorithm, natural science principles, for example) - include those as well.
  - You may use engineering tools (CAD, measuring tools, machine tools, for example). List those here.
  - Test requirement and methods/plan (what you need to test and how)

<Add more subsections for your other functional units>

- **Risk Analysis and Risk Management Plan**
  - Updated risk analysis and risk management plan (*review the risk analysis and risk management plan as documented in the User Requirement and Project Plan document and include an updated one in this design document*)

- **Appendices**
  - Datasheets of off-the-shelf components
  - New research, investigation and literature review.
  - Any other materials as appropriate

- **References:** See appendix B for the format

4.5. Presentation of Engineering Design

Deliverables:

- Oral presentation of engineering investigation, design and test plan
- Presentation material such as power-point document
- Demonstration of experimental and simulation work
- A document that describes the demonstration setup and highlights the key items of the demonstration (the purpose of the document is to convey what and how you are going to demonstrate)

Presentation and Demonstration Session

- 15 minutes oral presentation plus 5 minutes Q&A (question & answer) and 10 minutes for demonstration. The demonstration can be embedded in the presentation as long as the audience is not required to move to another place. Otherwise, the demonstration will happen after completion of all presentations.

- All capstone students must attend other students' presentations. They are also expected to ask questions during Q&A session.
- External guests may be invited and some of them may participate in the performance evaluation.
- Presentation and demonstration materials must be submitted in the designated eConestoga drop-box prior to the presentation.

Evaluation Criteria for the Presentation and Demonstration

- Communication skills (project related)
    - Each part of the presentation/demonstration should cover all major aspects as listed in the design document structure.
    - Quality of each part
- Communication skills (overall)
    - Well integrated and logically sequenced presentation/demonstration
    - Smooth presentation/demonstration flow, well-articulated answer to relevant questions, and good aural quality.
    - Clear presentation materials (power-point, audio and video, for example) – easy to read, see and hear
- Professionalism (overall): professional attire and behaviour consistent with professional work-ethic (punctuality, timeliness, responsibility, active participation and appropriate communication etiquette)

Presentation Structure/Sequence

The structure/sequence of the presentation is same as the structure of engineering design and test-plan document.

- Introduction (introduce yourself, state the purpose of the presentation, project title and definition)
- System Requirements
- System Design
- Major Functional Units under Design and Development
- Risks Analysis and Risk Management Plan

Presentation Guidelines

- The goal of this presentation is to establish that the engineering design is;
    - Substantial and at the level of 4th year engineering students
    - Founded on sound engineering investigations, experiments, simulations, analysis and appropriate engineering process of design decisions
    - To be verified with robust test plans
- The presentation makes a case that the design work is 4th year engineering level and volume of design work is appropriate for two semester long team project.
- The core of your presentation is the summary of key components of your Engineering Design and Test Plan document. Highlight the key design work and sample of few test plans, and be ready to explain/present further detail, if you are asked for it.
- **Quality of design**: give your best effort to highlight design complexity, depth and breadth of your research & investigation, precisely present your design with appropriate

means of engineering documentations (circuit-diagram/schematic for electronic hardware, flow-chart for software, mathematical formula for engineering computation and 3D graphics for mechanical components, for example). You should also highlight all design options and defend your design decisions.

4.5.1. Demonstration

Demonstration Structure/Sequence

- Describe the demonstration setup, highlight the key items of the demonstration, and explain how these experiments are useful to make engineering design decisions.
- Conduct demonstration in a way that the audience can clearly understand
- Make a case that the results of the experiments/simulations are the basis of some key engineering design decision.

Demonstration Guidelines

- Though you are not expected to implement any significant hardware or software at this stage, you are expected to perform mathematical analysis, experiment, simulation and/or test, as appropriate for the purpose of engineering design. You may also need some of those investigation work to select off-the-shelve components/functional units and open-source software. You will demonstrate these works along with the presentation.
- Be ready to demonstrate your experimental and simulation work. Prepare and display a document/drawing that describes the demonstration setup and highlights the key items of the demonstration (the purpose of the document is to convey what and how you are going to demonstrate) – only pointing to, for example, wires, circuit-board, PC screen or a part of program-code will not be enough.
- Give your best effort to prove that the experiments, simulations and/or tests are useful for your engineering decisions (design and/or off-the-shelve component selection).

4.6. Project Management, Team-Work and Professionalism

Project Deliverable:

- Weekly status report (see the template in Appendix D)
- Project binder

It is expected that each team will demonstrate skills of project management, team-work, and professionalism throughout the project duration. It is also expected that each individual student will carry out her/his assigned work and also be helpful to other team members.

Students will participate weekly debriefing and submit weekly status report for the purpose of assessing how well the current status of the project aligned with the original project plan. Each team will submit up-to-date project binder as well.

The faculty team will record students' project management and team-work performance as well as professionalism every week and mark the evaluation rubric in the last week of an evaluation cycle.

In order to do well in this part of the evaluation, you may consider the following guidelines:

- Be ready for debriefing with printed status report and project binder
- Be ready for a demo, if any (keep relevant document such as circuit/network/schematic diagram handy).
- Upload all documents in designated eConestoga drop-box before the debriefing unless there is any explicitly given submission deadline.
- Use debriefing time effectively.
- Team members should be on the same page. Each team member should know each other's work and status and be able to represent another member in her/his absence.
- Appropriate attire and communication etiquette.
- Never miss deadlines, and complete work 100%.
- Never miss attendance/participation. If deemed necessary, notify as early as possible with acceptable reason of absence.
- Your attitude and behavior must not have adverse impact on people around you in the college.
- Though you are in a project team, all evaluations, at least partially, are individual evaluations. Do your part well. Don't create any situation that might have negative impact on other team members' performance.
- Produce weekly project status report accurately
- Produce engineering log regularly and effectively
- Record meeting minutes/notes
- Keep the project binder up-to-date

5. Project Deliverables Description – Part 2 (Semester 8)

Each student shall use industry standard techniques and best practices for all development works (prototyping, fabrication and testing, for example. An engineering log-book (paper or electronic) shall be used to document all work including the investigation, design, development, and testing of all systems and subsystems. Project management tools will be used to control the project.

The capstone project is intended to emulate an industry project in its design and implementation lifecycle. Any documents described in this section must be submitted using the standard ESE format specified in Appendix A. The evaluation rubrics for the project deliverables are available in a separate document.

5.1. Updating User Requirements and Project Plan Document

Students will update their User Requirement and Project Plan for this semester in first week of the semester.

- The updated user requirements must be consistent with the Design and Test plan document (submitted at the end of capstone project part-1 in semester 7).
- Extend the project plan to cover the whole semester. The extended plan will match with the milestones of this part (see Section 3.2 for the milestones).
- Revise the other parts of the project plan, as needed.
- Highlight the changes and submit this updated document before the debriefing session.

5.2. Development and Testing

Deliverables:

- Demonstration of all functional units, which are designed by the team.
- Documented evidence of all significant verification testing, results and analysis of the results.

There is no specific structure suggested for these deliverables. However, it is expected that each team will present its work in a structured way so that the reader can understand easily. You may consider taking pictures and videos in addition to real-time demonstration. Refined form of some of these materials are expected to be used in the final report. For that reason, you may use the structures of the relevant sections of the final report document. These sections are: Documentary Evidence of System Development/Prototyping and Testing, and Results & Analysis.

During evaluation of your performance, you are expected to present/demonstrate/defend the following items for each individual functional unit, which you designed.

- Physical evidence of the prototype of the functional unit.
- Test setup (use a drawing) and associated design and test-plans.
- Documented test/verification results.
- Associated safety or societal impact, if any, and the mitigation method.
- Be thorough in your record keeping:
  - Pictures of the prototypes
  - Pictures/drawing of test setup, and videos of operation and testing
  - Test results and analysis
  - Engineering log and well-organized and up-to-date project binder

Note that, the faculty team will check-off a functional unit during one of the three assessment sessions (Assessment 1, 2 and 3 as indicated Section 3.2). Any completion after Assessment 3 will not be credited with any mark directly but will help to get better mark in system integration assessment.

5.3. System Integration and Performance Analysis

Deliverables:

- Demonstration of fully functional integrated system
- Documented evidence of all significant verification and validation testing, results and analysis of the results.

There is no specific structure suggested for these deliverables. However, it is expected that each team will present its work in a structured way so that the reader can understand easily. You may consider taking pictures and video in addition to real-time demonstration. Refined form of some of these materials are expected to be used in the final report. For that reason, you may use the structures of the relevant sections of the final report document. These sections are: Documentary Evidence of System Development/Prototyping and Testing, and Results & Analysis.

It is expected that all the designed functional units are successfully prototyped and tested before the beginning of this phase. Students will now integrate the functional units to build the complete system and perform verification test. Once the system found fully functional, the student will complete the validation test using SMART requirement (as defined in User Requirement document) as the benchmark.

During evaluation of your performance, you are expected to present/demonstrate/defend the following items:

- Physical evidence of the full system prototype.
- Test setup (use a drawing) and associated design and test-plans.
- Documented test/verification results.
- Associated safety or societal impact, if any, and the mitigation method.
- Be thorough in your record keeping:
  - Pictures of the system prototype
  - Pictures/drawing of test setup, and videos of operation and testing
  - Test results and analysis
  - Engineering log and well-organized and up-to-date project binder

5.4. Final Technical Report

Deliverable

A formal project report (Final Technical Report) to present the complete project work. This document will be preserved for future readers.

This document includes refined form of different sections of previous documents and also includes new materials. Major previous materials include:

- Literature review (from proposal document)
- User requirements (from User Requirement document)
- Design and test plans (from Engineering Design and Test-Plan document)

New material includes evidence of prototype development & testing, results and analysis. An abstract and conclusion section also to be written to give the document a complete and presentable form.

5.4.1. Structure of Final Technical Report

- **Cover Page**: include the document name 'Final Capstone Project Report', project title, student names & IDs, academic term/year, course title and code, program name and college name.
- **Table of Content** (include page number) and **Table of Figure**
- **Abstract**: write a summary of the key salient points in the project
- **Introduction**
    - Purpose of the document: identify the purpose and audience of this document.

    Note: The following three sub-sections (project definition, background and proposed solution) are expected to from the introduction section of your project proposal document

    - Project Definition: name and briefly describe the proposed project (*elaborate description to be covered in separate section*).
    - Background:
        - Problem/need statement: state what problem it solves.
        - Rationale: justify why this problem needs to be solved.
    - Proposed Solution
        - Describe the solution.
        - Indicate if there is any similar solution available in literature. If available, state how the proposed one is different. Highlight the benefits of the proposed solution.
        - Identify which part of the system the proposed project is focusing on.
- **Literature Review**

  Note: You may use selected elements of the literature review you performed in your proposal document

  Summarize your literature review with appropriate reference to authoritative source to report existing solutions that are comparable or alternative to what you proposed for this project. Authoritative source includes technical journal, magazines, industry white paper, and product description from original vendor. Literature review should address the following:

- Establish that the research/project topic is currently a topic of active research and development and has potential to be a practical solution to a real-world problem.
- Provide the readers of this document with some examples of research/project that are already reported in authoritative journals and technical magazines (this is a sort of environmental/boundary scan)
- Identify and describe the literatures that form the basis or starting point of this project work (these can be any combination of, for example, a project idea, research work, off-the-shelf components, industry white papers on 'coming soon' products or solutions, new regulations and open standards)
- Compare the proposed solution to existing solutions, if any.

- **Project Requirements**

  Note: you may fill in this part with the relevant content from the latest version of User Requirement document.

  - Specify the scope of the project using MoSCoW (**Mu**st, **S**hould, **Co**uld, **W**on't) template in appendix C. The scope should be appropriate for a two-semester-long project of a team of 4th year engineering students.
  - Specify a set of criteria that the proposed project will meet at its completion. The criteria should be appropriate for a two-semester-long project of a team of 4th year engineering students. Specify the criteria using SMART (**S**pecific, **M**easurable, **A**chievable, **R**ealistic, **T**imely) goal template, given in appendix C

- **System Design**

  Note: you may fill in this part with the relevant content from Engineering Design and Test-Plan document.

  - System level diagrams & drawings, principle of operation with brief descriptions of each functional unit, indication of what components are 'off-the-shelf' and what are to be designed and developed.
  - Test plans for:
    - verification of system integration
    - validation test to confirm that the system meets the SMART requirement (as defined in User Requirement and Project Plan document)
  - Safety, societal and environmental issues associated with the project and a mitigation plan.
    - Describe how your solution may impact on the environment and society (positive or negative). Describe how to reduce the negative impact, if any.
    - Describe required safety precautions if the product may cause any safety hazard.
  - Brief description/specification of off-the-shelf components (datasheet or detail description of off-the-shelf components such as microcontroller, wireless transceivers, embedded modules, major electrical & electronic components, communication devices, client-server software, open-source software and protocols should be included as an appendix, unless it is too large. If too big, those should be submitted as separate files).

- **Design of Functional Units** (multiple sections with appropriate titles – one for each functional unit)

  Note: you may fill in this part with the relevant content from Engineering Design and Test-Plan document.

  You may create one subsection per functional unit to present your design work. The following sub-section is an example.

  - Definition of the functional unit.
  - Principle of operation, schematics, flowchart, pseudocode and/or any other engineering drawing, as appropriate.
  - Design specification
  - Test requirements and methods/plan

<Add more subsections for your other functional units>

- **Documentary Evidence of System Development/Prototyping and Testing**

  You recorded evidence during prototyping, testing and system integration in the following two phases:

  - Development and Testing
  - System Integration and Performance Analysis

  Refine those evidence and add here

  Example:

  - Photograph of hardware you implemented and its experimental setup
  - PCB board you developed
  - Link to a video clip that demonstrate the system operation and testing
  - Reference to software code you developed (should be placed as an appendix if small, or separate files if large)

- **Results and Analysis**
  - Qualitative Analysis: Present your qualitative test results here. It is expected that you will present results under different test conditions.
  - Test Results: Collect test data and present those data here. Data tables are expected here.
  - Quantitative Analysis: Analyze the data to present key characteristic features. Graph of different kinds may be appropriate here.

- **Conclusion and Recommendations**
  - Comment on the project success, strength & weakness, learning experience, for example.
  - Make some recommendation for future students who may undertake similar project or want to continue the same project. What can be done in future and what should be avoided, for example.

- **Appendices:** Add additional materials that you may have but not included in the main body of the report. Here are some examples:
  - Software code that you have written
  - Specifications of off-the-shelf components and equipment

    - Information about APIs and open-source code you used in your project
    - Drawings and schematic diagrams

- **References and Bibliography:** See Appendix B for the format

5.5. Final Technical Presentation and Demonstration

Deliverables:

- Oral presentation
- Presentation material such as power-point document
- Demonstration of completely integrated system
- A document that describes the demonstration setup and highlights the key items of the demonstration (the purpose of the document is to convey what and how you are going to demonstrate)
- A video of the project work

Each team will present and demonstrate its final form of the project work. ESE students, non-capstone faculty members and staff may attend. Representatives of project sponsors and other interested professional and technical people may attend as well. Some of them may participate in performance evaluation.

Presentation and Demonstration Session

- 20 minutes oral presentation plus 5 minutes Q&A (question & answer) and 10 minutes for demonstration. The demonstration can be embedded in the presentation as long as the audience is not required to move to another place. Otherwise, the demonstration will happen after completion of all presentations.
- All capstone students must attend other students' presentations. They are also expected to ask questions during Q&A session.
- External guests may be invited and some of them may participate in the performance evaluation.
- Presentation and demonstration materials must be submitted in the designated eConestoga drop-box prior to the presentation.

Evaluation Criteria for the Presentation and Demonstration

- Communication skills (project related)
    - Each part of the presentation/demonstration should cover all major aspects as listed in the structure of Final Capstone Project Report document.
    - Quality of each part
- Communication skills (overall)
    - Well integrated and logically sequenced presentation/demonstration
    - Smooth presentation/demonstration flow, well-articulated answer to relevant questions, and good aural quality.
    - Clear presentation materials (power-point, audio and video, for example) – easy to read, see and hear

- Professionalism (overall): professional attire and behaviour consistent with professional work-ethic (punctuality, timeliness, responsibility, active participation and appropriate communication etiquette)

Presentation Structure/Sequence

The structure/sequence of the presentation is same as the structure of the final report document.

- Introduction (introduce yourself, state the purpose of the presentation, project title and definition & description)
- System Requirements
- System Design and Test Plans
- Design of Major Functional Units and Test Plans
- Evidence of Prototyping and Testing
- 2 to 3 minutes long video clip of elevator pitch covering the following aspects:
  - the problem the project tried to solve
  - the solution detail highlighting innovative aspects of the solution
  - future directions for the project, specifically commercialization potential

  (*an elevator pitch is a brief, persuasive speech that you use to spark interest in your project work*).
- Results and Analysis
- Recommendations and Conclusion

Presentation Guidelines

- The goal of this presentation is to establish the following points:
  - The project involves substantial and at the level of 4th year engineering students
  - The project is founded on sound engineering investigations, experiments, simulations, analysis and appropriate engineering process of design decisions
  - Prototyped as per design and MoSCoW scope
  - Verified and validated with robust test plans
  - Performed performance analysis of the integrated system in comparison to the proposed SMART goals.
- Make the presentation suitable for the whole audience and demonstrate good communication skills.

5.5.1. Demonstration

Demonstration Structure/Sequence

- Describe the demonstration setup and highlights the key items of the demonstration and explain how well prototype meets the proposed goals.
- Conduct demonstration in a way that the audience can clearly understand.
- Be ready to demonstrate. Prepare and display a document/drawing that describes the demonstration setup and highlights the key items of the demonstration (the purpose of the document is to convey what and how you are going to demonstrate) – only pointing to, for example, wires, circuit-board, PC screen or a part of program-code will not be enough.

5.6. Project Management and Professionalism

See section 4.6

5.7. Oral Review and Online Survey

The oral review is an interview where the students will share their project experience with the faculty team. The theme of the review includes (but not necessarily limited to) the following aspects:

- Students will identify strength and weaknesses of their project work (the most as well as the least successful outcomes, what went well and what not, what they would do differently if they would have another chance and so on).
- Share their professional opinion on the course delivery (what they liked and where the faculty team should work on to improve, quality and effectiveness of the project guideline document and rubrics, and what should be done to improve these documents, and so on)

The students will take an on-line survey in eConestoga to express their opinion on Capstone project at the end of semester.

5.8. Tech Showcase

Deliverables: a video of the project work

Only one project will be nominated from ESE program for Technical Showcase event and Mastercraft Award. Normally (though not necessarily) the best capstone project gets the nomination. However, it is possible that another project is selected if it has higher potential to meet the criteria of the Mastercraft award better.

Selection Criteria of the Mastercraft Award: The participating project will be evaluated by the Judges of Mastercraft Award (not ESE faculty team) based on the following criteria: first impression, innovation, creativity, skills and quality.

Appendix A: ESE Document Format

Each technical report must be word processed and checked for spelling and grammar. It is to be 1.5 line-spaced using 12-point Times New Roman or 12-point Calibri or 10-point Verdana font.

Pages in the Front Matter are numbered using Roman numerals. The Title Page is considered page "i" but is not numbered. The Abstract is not included in the page numbering at all. The Table of Contents is the second page of the Front Matter and is numbered with a small Roman numeral "ii" centred in the footer at the bottom of the page. All subsequent Front Matter pages are numbered in a similar way.

Pages in the Main Body of the report are numbered using Arabic numbers. The "1" for the first page is centred in the footer at the bottom of the page. All subsequent Main Body pages are numbered in the top right corner of the page.

Pages in the Back Matter are numbered in the top right corner of the page and continue on from the number assigned to the last page of the Main Body.

Headings must be used to identify the sections within the technical report. As in this document, headings in the technical report are to be outline numbered, with each major section getting a new number.

All figures and tables should be referenced in the text at the relevant point. The reference should precede the graphic which is normally presented along with the text. All graphics must be numbered, titled and referenced appropriately. All illustrations, drawings, maps, graphs and charts are considered figures and should be included in the list of figures. Place the number and title below the figure. All tables should be included in the list of tables. Place the number and title above the table.

Avoid typographical anomalies like "orphans" and "widows". Orphans are short sections (usually single lines or headings) that belong with the text on the following page. Widows are single lines or words that are isolated at the top of a page but belong to the text on the previous page.

Appendix B: Reference and Bibliography

Template with examples:

[1] J. Yick, et al., "Wireless sensor network survey," Computer Networks, vol. 52, pp. 2292-2330, 2008.

[2] Y. E. Krasteva, et al., "Remote HW-SW reconfigurable Wireless Sensor nodes," in Industrial Electronics, 2008. IECON 2008. 34th Annual Conference of IEEE, 2008, pp. 24832488.

[3] J. Jones. (1991, May 10). Networks (2nd ed.) [Online access on June 14, 2016]. URL: http://www.atm.com,

For more examples see: http://www.ieee.org/documents/ieeecitationref.pdf

Note: You may use the built-in bibliography functionality in Microsoft WORD. We recommend the IEEE 2006 Style.

Appendix C: MoSCoW and SMART Templates

MoSCoW Table (list the requirements and select its category with an 'X')

Requirements	Must	Should	Could	Wouldn't
Requirement 1	X	-	-	-
Requirement 2	-	X	-	-
Requirement 2	-	-	X	-

SMART Table (fill in the right column for each task)

Task/Goal Title	<write the title here>
Specific	<answer questions such as: what exactly the task is? who will do this? What resources/knowledge it requires?>
Measurable	<answer questions such as: how to measure the success? what is the benchmark to measure the success?>
Attainable	<answer questions such as: how is it realistic to attain? what are the main obstacles, if any?>
Relevant	<answer questions such as: how is it useful for the project?>
Time-Bound	<answer questions such as: when it needs to be finished? how other tasks depend on this task and vice versa?>

Appendix D: Project Status Templates

Project Status Report – Week #

2026-MM-DD

Project Name: **Team Members**:

Project Overall Status: GREEN/YELLOW/RED

Status Overview:

Activity Title with start date and also planned completion date and name of responsible team member	**Status** (Last Week) G/Y/R	**Status** (This Week)		**Plan** (Next Week)
		Plan (as planned last week for this week)	**Actual** Green/Yellow/Red	

- Overall Status: Green is the best, Red is the worst, and Yellow is in between. You may add + or − suffix to indicate the status more accurately.
- Planned completion date and name of responsible team member: indicate if these are unchanged or changed/new.
- You have to compare your status in three ways:
    1. Status of this week versus the status in last week
    2. Status of this week versus the plan for this week
    3. Planned completion date and responsible person (this week) versus the planned completion date and responsible person (last week)

Appendix E: Graduate Attributes

Complete List

https://engineerscanada.ca/sites/default/files/Accreditation_Criteria_Procedures_2015.pdf

3.1.1 **A knowledge base for engineering:** Demonstrated competence in university level mathematics, natural sciences, engineering fundamentals, and specialized engineering knowledge appropriate to the program.

3.1.2 **Problem analysis:** An ability to use appropriate knowledge and skills to identify, formulate, analyze, and solve complex engineering problems in order to reach substantiated conclusions.

3.1.3 **Investigation:** An ability to conduct investigations of complex problems by methods that include appropriate experiments, analysis and interpretation of data and synthesis of information in order to reach valid conclusions.

3.1.4 **Design:** An ability to design solutions for complex, open-ended engineering problems and to design systems, components or processes that meet specified needs with appropriate attention to health and safety risks, applicable standards, and economic, environmental, cultural and societal considerations.

3.1.5 **Use of engineering tools:** An ability to create, select, apply, adapt, and extend appropriate techniques, resources, and modern engineering tools to a range of engineering activities, from simple to complex, with an understanding of the associated limitations.

3.1.6 **Individual and team work:** An ability to work effectively as a member and leader in teams, preferably in a multi-disciplinary setting.

3.1.7 **Communication skills:** An ability to communicate complex engineering concepts within the profession and with society at large. Such ability includes reading, writing, speaking and listening, and the ability to comprehend and write effective reports and design documentation, and to give and effectively respond to clear instructions.

3.1.8 **Professionalism:** An understanding of the roles and responsibilities of the professional engineer in society, especially the primary role of protection of the public and the public interest.

3.1.9 **Impact of engineering on society and the environment:** An ability to analyze social and environmental aspects of engineering activities. Such ability includes an understanding of the interactions that engineering has with the economic, social, health, safety, legal, and cultural aspects of society, the uncertainties in the prediction of such interactions; and the concepts of sustainable design and development and environmental stewardship.

3.1.10 **Ethics and equity:** An ability to apply professional ethics, accountability, and equity.

3.1.11 **Economics and project management:** An ability to appropriately incorporate economics and business practices including project, risk, and change management into the practice of engineering and to understand their limitations.

3.1.12 **Life-long learning:** An ability to identify and to address their own educational needs in a changing world in ways sufficient to maintain their competence and to allow them to contribute to the advancement of knowledge.

The Capstone Grad Attributes

Professional Body of Knowledge

Code	Attribute	Level
KB	**Knowledge base**	
KB1	Math	
KB2	Natural Science	
KB3	Fundamental Engineering	
KB4	Specialized Engineering	
KB5	Complementary Studies	
PA	**Problem analysis**	
PA1	Decomposition	
PA2	Methodology	
PA3	Validation	
IV	**Investigation**	
IV1	Research	A
IV2	Measure	A
IV3	Experiment	A
	Design	
ED1	Problem Design & Research	A
ED2	Preliminary Design	A
ED3	Detailed Design	A
ED4	Implementation	A
ED5	Verification & Validation	A
ET	**Use of engineering tools**	
ET1	Models/Simulations	A
ET2	Measurement Tools	A
ET3	Manufacturing Tools	
ET4	CAD Systems	A

Employability Skills

Code	Attribute	Level
TM	**Individual and team work**	
TM1	Personal Contribution	
TM2	Collaboration	
TM3	Infrastructure	
CM	**Communication skills**	
CM1	Log Engineering Info	A
CM2	Convey Engineering Info	A
PR	**Professionalism**	
PR1	Work Ethic	A
PR2	Professional Conduct	
PR3	Professional Contribution	
PR4	Professional Practice	
LL	**Life-long learning**	
LL1	Autonomous Learning	D
LL2	Applying Knowledge & Skills	D
LL3	Self Direction & Reflection (Metalearning)	A
LL4	Learning Strategies (Metacognition)	

Professional Responsibility

Code	Attribute	Level
SC	**Impact on society and the environment**	
SC1	Environmental Awareness	A
SC2	Product Life Cycle	
SC3	Balance & Tradeoff	
EE	**Ethics and equity**	
EE1	Ethical Responsibility	A
PM	**Economics and project management**	
PM1	Project Scheduling	A
PM2	Resource Allocation & Costing	A
PM3	Risk Management	A
PM4	Business Planning	
PM5	Economic Analysis	

LEGEND		
I	**Introduced**	first experience/use
D	**Developed**	continued experience/use
A	**Applied**	integration/extension of knowledge & skills